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Examen Simulacro - Intensivo 2024/2025

- A scientist mixes 75 ml of Solution A with 50 ml of Solution X. The first solution has 40% hydrochloric acid concentration, the second has an unknown hydrochloric acid concentration. When the scientist tests the resulting 125 ml solution, he finds that it has 36% hydrochloric acid concentration. What is the hydrochloric acid concentration of Solution X?

(A) 90 % (B) 55% (C) 30% (D) 76% (E) 50%
- A polynomial is given $P(x) = -2x^7 - 10x^6 + 8x^5 + 40x^4$, what are the zeros of the polynomial?

(A) 0, ± 2 , ± 5 , ± 1 (B) ± 2 , ± 5 , ± 1 (C) 0, ± 2 , ± 5
 (D) 0, ± 2 , -5 (E) 0, ± 2 , ± 5 , ± 4
- Which one of the given functions are NOT inverses of each other?

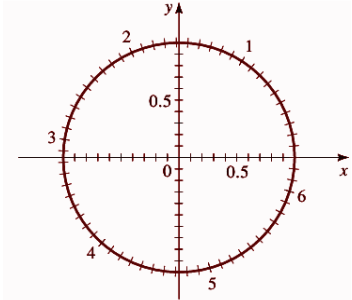
(A) $g(x) = -x^5 - 3$ $f(x) = \sqrt[5]{-x - 3}$ (B) $g(x) = \frac{-x-1}{x-2}$ $f(x) = \frac{-2x+1}{-x-1}$
 (C) $g(x) = -\frac{2}{x+3}$ $f(x) = \frac{3x+2}{x+2}$ (D) $g(x) = \frac{x-5}{10}$ $h(x) = 10x + 5$
 (E) None of the above
- A function is given $y = 2 - 5(x + 4)^2$, indicate the transformations of $f(x) = x^2$ that were applied to its graph in the correct order?

(A) Shifted 4 units to the left, reflected over the x-axis and stretched horizontally by a factor of 5, and shifted downward 2 units.
 (B) Shifted 4 units to the left, reflected over the y-axis and stretched vertically by a factor of 5, and shifted upward 2 units.
 (C) Shifted 4 units to the right, reflected over the x-axis and stretched horizontally by a factor of 5, and shifted downward 2 units.
 (D) Shifted 4 units to the left, reflected over the x-axis and stretched vertically by a factor of 5, and shifted upward 2 units.
 (E) None of the above.
- A rational function is defined as: $r(x) = \frac{x^2+6x-4}{3x-6}$
 What is the slant asymptote of $r(x)$?

(A) $y = \frac{1}{3}x + \frac{8}{3}$ (B) $y = \frac{x^2}{3}$ (C) $y = 3x - 6$ (D) $y = \frac{1}{3}x + \frac{1}{3}$ (E) none of the above
- Two siblings have a combined annual income of \$101,250. One sibling earns 25% more than the other. What is the annual income of the sibling earning more?

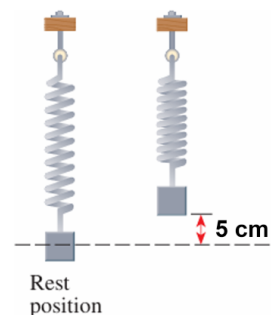
(A) 45,000 (B) 81,000 (C) 25,312 (D) 75,938 (E) 56,250
- In a cherry orchard, the number of cherries produced by each tree depends on how densely the trees are planted. If n trees are planted on an acre of land, then each tree produces $800 - 8n$ cherries. The total number of cherries produced per acre is given by: $A(n) = n(800 - 8n)$. How many trees should be planted per acre to obtain the maximum yield of cherries?

(A) 50,000 (B) 20,000 (C) 500 (D) 50 (E) 20

8. If there are three functions, such as $f(x) = \frac{x}{2}$, $g(x) = 2x - 1$ and $h(x) = 3x^2$. Then find the composition of these functions such as $[f \circ (g \circ h)](x)$ for $x = -1$.
 (A) 2.5 (B) -1 (C) 2 (D) -0.5 (E) 1
9. The slope of the sides of a pyramid in South America is $\frac{7}{5}$. If the base of the pyramid is 600 ft, how tall is the pyramid?
 (A) 525 ft (B) 360 ft (C) 420 ft (D) 700 ft (E) 380 ft
10. Find an equation of the line that passes through the point $(-1, 4)$ and is perpendicular to the line passing through the points $(2, 3)$ and $(6, 7)$.
 (A) $y = x + 3$ (B) $y = -x + 3$ (C) $y = -x - 3$ (D) $y = -x + 4$ (E) none of the above
11. A sample of *Polonium-210* has a *half-life* of 140 days. Suppose a sample of this substance has a mass of 300 g. Find the mass remaining after 280 days.
 (A) 75 g (B) 35.5 g (C) 37.5 g (D) 150 g (E) 57.5 g
12. In 2024 UPTP students in Taiwan experienced the largest earthquake in 25 years with a magnitude of 7.4 in the *Richter Scale*. Find the intensity I of the earthquake in *centimeters*. (Hint: $10^{0.4} = 2.5$; $s = 10^{-4}$).
 (A) 5000 (B) 2400 (C) 2800 (D) 2500 (E) 2750
13. Use the laws of logarithms to expand the expression $\ln\left(\frac{ab}{\sqrt[3]{c}}\right)$
 (A) $\ln(a) + \ln(b) - \frac{1}{3}\ln(c)$ (B) $\ln(a) - \ln(b) + \frac{1}{3}\ln(c)$
 (C) $\ln(a) - \ln(b) + 3\ln(c)$ (D) $\ln(a) + \ln(b) - 3\ln(c)$
 (E) None of the above
14. Find the exact value of the evaluated trigonometric function $\sin\left(-\frac{7\pi}{6}\right)$.
 a) $\frac{1}{2}$ b) $\frac{\sqrt{3}}{2}$ c) -1 d) $-\frac{\sqrt{3}}{2}$ e) $-\frac{1}{2}$
15. Find the approximate value of the trigonometric function $\sin(1.198)$ by using the following figure.
 (A) 0.51
 (B) 0.8
 (C) 0.91
 (D) 1
 (E) 1.1
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16. Which of the following trigonometric functions is *negative* in the third quadrant?
 (A) $\sin(\theta)$ (B) $\sec(\theta)$ (C) $\cot(\theta)$ (D) A & B are correct
 (E) A & C are correct.
17. Which of the following is an ODD function?
 (A) $2^{\sin(x)}$ (B) $x^5 + |x|$ (C) $\tan(x) + \cos(x)$ (D) $\sin(x) + x^3$ (E) e^x

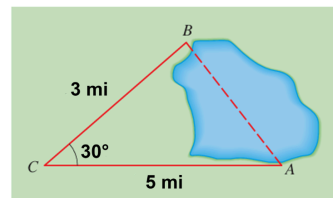
18. A spring in simple harmonic motion is compressed 5 cm and then released. It is observed that return to the compressed position in $\frac{1}{4}$ sg. Which of the following functions represents its motion?

(A) $y = \frac{1}{4} \cos(5\pi t)$
 (B) $y = 5 \sin(4\pi t)$
 (C) $y = \frac{1}{4} \sin(5\pi t)$
 (D) $y = 5 \cos(8\pi t)$
 (E) None of the above



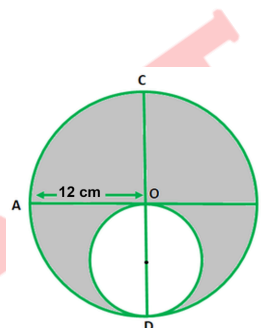
19. To find the distance across a small lake, a surveyor has taken the measurements shown. Find the distance across the lake using this information.

(A) $\sqrt{19} \text{ mi}$ (B) $\sqrt{34 - 30\sqrt{2}} \text{ mi}$
 (C) $\sqrt{34 - (15\sqrt{3})} \text{ mi}$ (D) $\sqrt{34\sqrt{3}} \text{ mi}$
 (E) $\sqrt{34 - \sqrt{3}} \text{ mi}$



20. Find the shaded area.

(A) 128π
 (B) 135π
 (C) 576π
 (D) 108π
 (E) None of the above



21. Simplify the trigonometric expression $\frac{\sin(x+\pi) \cdot \tan(x-\frac{\pi}{2}) \cdot \cos(\pi-x)}{\tan(x+\pi) \cdot \sin(\frac{\pi}{2}-x) \cdot \sec(\frac{\pi}{2}-x)} + \frac{\sin(2x)}{(\sin x + \cos x)^2 - 1}$, where $0 < x < \frac{\pi}{2}$.

(A) $\cos^2 x$ (B) $\sin^2 x$ (C) 1 (D) $1 - \sin x$ (E) $\cos x - 1$

22. Which of the following options is NOT a solution for the trigonometric equation $\cos(3x) + \cos(2x) + \cos(x) = 0$

(A) $\frac{5\pi}{4}$ (B) $\frac{3\pi}{4}$ (C) $\frac{5\pi}{3}$ (D) $\frac{4\pi}{3}$ (E) $\frac{10\pi}{3}$

23. Determine the value of $\cos[\tan^{-1}(\frac{4}{3}) + 2\sin^{-1}(\frac{2}{3})]$

(A) $\frac{3-5\sqrt{3}}{45}$ (B) $\frac{1-5\sqrt{5}}{15}$ (C) $\frac{3-16\sqrt{2}}{15}$ (D) $\frac{1-16\sqrt{3}}{45}$ (E) $\frac{3-16\sqrt{5}}{45}$

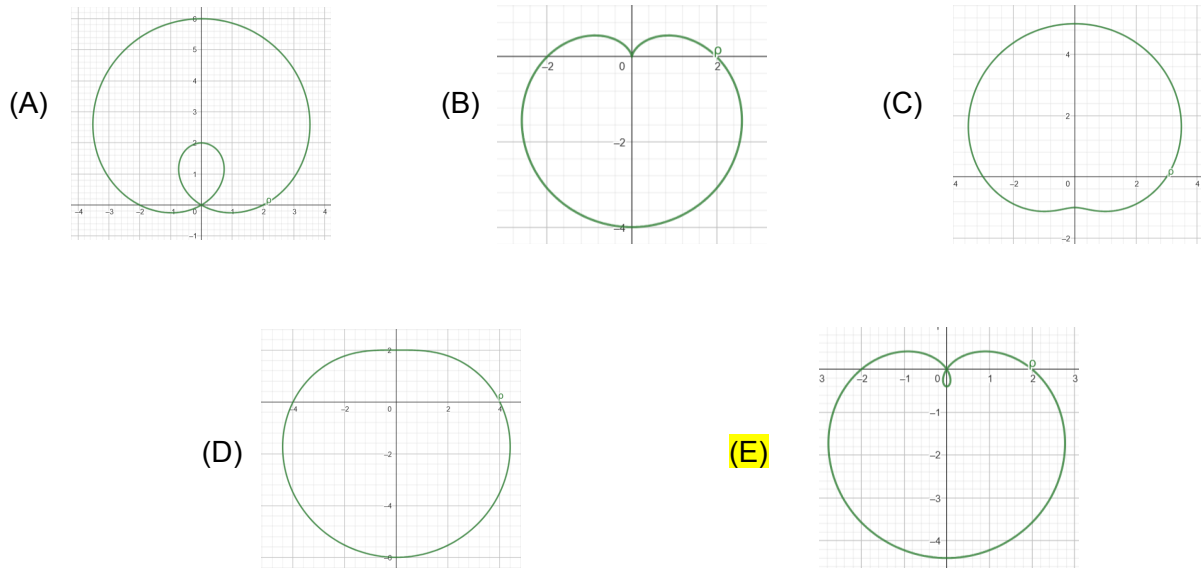
24. Determine the value of $\sin(\frac{7\pi}{12})$

(A) $\frac{\sqrt{3}+\sqrt{6}}{2}$ (B) $\frac{\sqrt{6}+\sqrt{3}}{4}$ (C) $\frac{\sqrt{3}+\sqrt{2}}{2}$ (D) $\frac{\sqrt{6}+\sqrt{2}}{4}$
 (E) None of the above

25. Find the value of $(-3 + \sqrt{3}i)^4$

(A) $-72 - 72\sqrt{3}i$ (B) $-36\sqrt{2} + 36\sqrt{2}i$ (C) $-36\sqrt{3} - 36i$ (D) $72\sqrt{3} + 72i$
 (E) None of the above

26. Which of the following graphs is given by the polar equation $r = 2 - \frac{5}{2}\sin\theta$



27. The parametric equations $x = \sqrt{2}\cos(t)$ $y = 3\sin(t)$ forms a graph that is best described by which of the following options:

- (A) A circle with center at $(\sqrt{2}, 3)$
 (B) An ellipse with major axis of $2\sqrt{2}$
 (C) A parabola with focus at $(\sqrt{2}, 3)$
 (D) An ellipse with an eccentricity of $\frac{\sqrt{7}}{3}$
 (E) A hyperbola with vertices at $(\sqrt{2}, \pm 3)$

28. A migrating salmon heads in the direction $N 30^\circ E$, swimming at 5 mi/h relative to the water. The prevailing ocean currents flow due east at 3 mi/h . Find the true speed and direction of the salmon.

- (A) $V = 7 \text{ mi/h}$, $\theta = \tan^{-1}(\frac{11\sqrt{3}}{15})$ (B) $V = 7.77 \text{ mi/h}$, $\theta = \tan^{-1}(\frac{11\sqrt{3}}{15})$
 (C) $V = 7 \text{ mi/h}$, $\theta = \tan^{-1}(\frac{5\sqrt{3}}{11})$ (D) $V = 7.77 \text{ mi/h}$, $\theta = \tan^{-1}(\frac{5\sqrt{3}}{11})$
 (E) None of the above

29. Find the area of the circle formed when the plane $y - 3 = 0$ intercepts the sphere

$$x^2 + y^2 + z^2 - 4x - 2z + 2y - 30 = 0$$

- (A) 16π (B) 20π (C) 32π (D) 6π (E) 4π

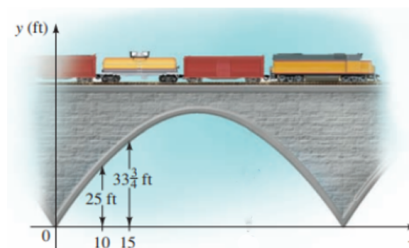
30. Find the equation of the plane that passes through the points

$P(2, 0, -6)$, $Q(0, -4, -6)$ and $R(5, 3, 0)$. Notice that the equation of the plane is in the form of $x + ay + bz = K$.

- (A) $x - 2y - 4z = \frac{7}{2}$ (B) $x + \frac{1}{2}y - \frac{1}{4}z = 7$ (C) $x - \frac{1}{4}y - \frac{1}{2}z = 7$
 (D) $x - \frac{1}{2}y - \frac{1}{4}z = \frac{7}{2}$ (E) $x + \frac{1}{4}y - \frac{1}{2}z = 14$

31. The opening of a railway bridge over a roadway is in the shape of a parabola. A surveyor measures the heights of three points on the bridge, as shown in the figure. He wishes to find an equation of the form $y = ax^2 + bx + c$. Use the surveyed points to set up a system of linear equations to find the values of the coefficients a , b and c .

- (A) $a = -\frac{1}{5}$, $b = 2$, $c = 25$
 (B) $a = 1$, $b = -9$, $c = 10$
 (C) $a = -\frac{1}{4}$, $b = 10$, $c = 0$
 (D) $a = -\frac{1}{20}$, $b = 3$, $c = 0$
 (E) $a = -\frac{1}{25}$, $b = \frac{1}{3}$, $c = -6$



32. Given the following system of equations $\begin{cases} x + y = 1 \\ y + z = 2 \\ z + w = 3 \\ w - x = 4 \end{cases}$ use Cramer's Rule to find the value of $\frac{|D_x|}{|D_y|}$

- (A) $\frac{1}{5}$ (B) $\frac{1}{2}$ (C) $-\frac{1}{2}$ (D) $-\frac{1}{5}$ (E) $\frac{1}{4}$

33. Find the area of the triangle with vertices at $(-2, 3)$, $(1, 4)$ and $(2, -1)$.

- (A) 16 (B) 32 (C) 8 (D) 4 (E) 2

34. Find the value of A in the partial fraction decomposition of the rational function

$$\frac{x^2 + x - 2}{x^3 + x^2 - 9x - 9} = \frac{A}{(x+1)} + \frac{1}{3(x+3)} + \frac{5}{12(x-3)}$$

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) $\frac{1}{6}$ (D) $-\frac{1}{5}$ (E) $\frac{1}{5}$

35. Identify the conic determined by the equation $4x^2 + 7xy + 8y^2 + 36x - 48y = 12$

- (A) Parabola (B) Hyperbola (C) Ellipse (D) Circle (E) Two straight lines

36. A polar equation of a conic is given by $r = \frac{12}{5-a \cos(\theta)}$, find the value of a so the equation represents a hyperbola.

- (A) 1 (B) 7 (C) 5 (D) 4 (E) 2

37. The expression $2 + 1 + \dots + \frac{100}{197}$ is equivalent to

- (A) $\sum_{k=1}^{99} \frac{k+3}{k+1}$ (B) $\sum_{k=0}^{98} \frac{k+2}{2k+1}$ (C) $\sum_{k=-1}^{101} \frac{k-1}{2k+1}$ (D) $\sum_{k=0}^{100} \frac{k(k-1)}{k+1}$ (E) $\sum_{k=3}^{100} k - 1$

38. The n th term of an arithmetic sequences is given by $a_n = a + (n - 1)d$. Find the sum of the first 41 numbers if $a = 3$ and $d = 4$.

- (A) 3403 (B) 3551 (C) 3482 (D) 3731 (E) 3039

39. Identify the geometric sequence out of the following options

- (A) $a_n = \frac{n(n+1)}{2}$ (B) $a_{n+1} = (a_n)(a_{n-1})$ (C) $a_n = \frac{5}{2^n} + 1$ (D) $a_n = 3^n + 2^n$
 (E) $a_n = (a_{n-1})(2)$

40. The following is a well-known children's rhyme:

As I was going to St. Ives,
I met a man with seven wives;
Every wife had seven children;
Every children had seven cats;
Every cat had seven kits;
Kits, cats, children, and wives,
How many were going to St. Ives?

Assuming that the entire group is actually going to St. Ives, find the sum.

- (A) 2905 (B) 2058 (C) 2791 (D) 934 (E) 2801

41. Evaluate $\lim_{t \rightarrow 4} \frac{\frac{1}{\sqrt{t}} - \frac{1}{2}}{t-4}$.

- (A) 1 (B) $-\frac{1}{16}$ (C) $\frac{1}{16}$ (D) 16 (E) -16

42. Does the limit $\lim_{t \rightarrow 1.5} \frac{2x^2-3x}{|2x-3|}$ exist? If it does, what's the value?

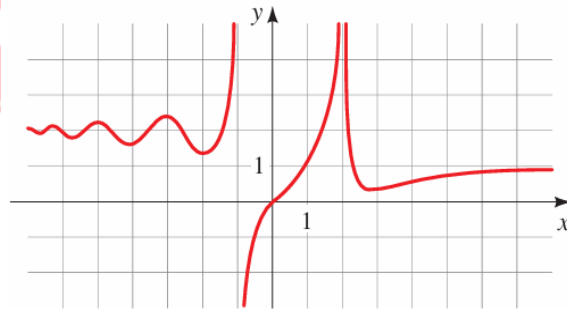
- (A) Yes, and its equal to 0 (B) Yes, and its equal to 1.5 (C) No, it does not exist
(D) Yes, and its equal to 3 (E) Yes, and its equal to 4

43. Find the derivative of the function $f(x) = 6x^2 + 4x - 2$ at the number 3

- (A) 40 (B) 28 (C) 38 (D) 23 (E) 39

44. Which of the following options is FALSE about the following function:

- (A) $\lim_{x \rightarrow -\infty} f(x) = 2$
(B) $\lim_{x \rightarrow \infty} f(x) = 1$
(C) $\lim_{x \rightarrow -1^+} f(x) = \infty$
(D) $\lim_{x \rightarrow 1} f(x) = 1$
(E) $\lim_{x \rightarrow 2^-} f(x) = \infty$



45. In order to find the area that lies under the curve $y = x^3$ where $0 \leq x \leq 5$ we divide it into approximating rectangles. Which of the following statements will be INCORRECT?

- (A) Width: $\frac{5}{n}$
(B) Right endpoint: $\frac{5k}{n}$
(C) Height: $\frac{125k^3}{n^3}$
(D) The area under $y = x$ in the same interval would be LESS.
(E) All of the above are correct